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Reduced-Form Versus Structural Modeling in Environmental and Resource Economics

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Abstract

We contrast structural and reduced form empirical studies in environmental and resource economics. Both methodologies have their own context-specific advantages and disadvantages, and should be viewed as complements, not substitutes. Structural models typically require a theoretical model and explicit assumptions about structural errors in order to recover the parameters of behavioral functions. These estimates may be required to measure general equilibrium welfare effects or to simulate intricate feedback loops between natural and economic processes. However, many of the assumptions used to recover structural estimates are untestable. The goal of reduced form studies is, conversely, to recover key parameters of interest using exogenous within-sample variation with as few structural assumptions as possible—reducing reliance on these assumptions assists in establishing causality in the relationship of interest. Reduced-form studies do, however, require assumptions of their own, e.g., the (quasi) randomness of an experiment with no spillover effects on the control group.

1. INTRODUCTION

There is an active debate over the advantages and disadvantages of reduced-form versus structural approaches to empirical modeling. While some subfields of economics rely heavily on structural methods (e.g., industrial organization and, more recently, public finance), others tend toward more reduced-form approaches (e.g., labor and development economics). In this article, we review some of the primary issues that arise in the implementation of each methodological approach and discuss the advantages and disadvantages of each. In so doing, we focus our attention on recent examples in environmental and resource economics.

We feel both methodological approaches have important advantages and disadvantages and that each may be better suited to answering particular types of questions. On the one hand, the structural approach is particularly useful in the presence of feedback loops, or when rational agents can reoptimize in response to large policy shifts. Feedback loops pose difficult endogeneity problems that generally make them more difficult to estimate using reduced-form methods. Counter-factual policy simulations that include out-of-sample predictions by definition require structural parameters that model how people readjust in response to the policy.

Reduced-form studies, on the other hand, are better suited to identifying key high-level response parameters as opposed to primitives in a structural model. Causality and identification are established by focusing on the source of variation, while avoiding reliance on untestable functional-form, distributional, and other modeling assumptions, even though it should be noted that reduced-form studies do require assumptions of their own (Smith 2007).

Each approach therefore fills an important niche. One method may be preferable to the other for a particular empirical question, and the two methods may even be used in conjunction with one another to provide different perspectives on the same question. We would encourage the reader to view the methods as complements rather than as substitutes. For example, reduced-form studies can be used to estimate high-level response parameters of interest, e.g., the effect of environmental amenities (clean air, good weather, etc) on humans (e.g., infant health) or the ecosystem (e.g., crop yields) that are then used in structural models to simulate policy responses by economic agents. Recent work by Chetty (2008) highlights these complementarities.

Before beginning with our review, we start with a few disclaimers. First, we do not attempt a comprehensive review of either literature. Rather, the goal of this article is to highlight the strengths and weaknesses of both structural and reduced-form approaches by discussing a carefully selected set of empirical examples. Second, we do not attempt to describe exhaustively each of the papers that we do cite; instead, we focus our attention on the elements of each paper that are most relevant for explaining the strength or weakness in question. Third, although many of the papers are quite technical (especially those using structural methods), we do not use any math in our exposition. There is really no way to do so in a limited way—even with the short list of papers that we consider, describing the technical details of each would unduly extend this article. Instead, we hope that the description we provide will lead interested readers to explore these papers in more depth. We believe that the papers we have chosen provide good references for those interested in implementing these techniques.

The remainder of this article first discusses structural modeling techniques (Section 2) and then moves on to a description of reduced-form methods (Section 3). The discussion

of each method begins with a description of the ways in which it may prove useful to environmental economists. That description is then followed by a more detailed accounting of specific papers that illustrate those advantages. Section 4 provides a brief conclusion.

2. STRUCTURAL MODELS IN ENVIRONMENTAL AND RESOURCE ECONOMICS

In this section, we focus on the role of structural models in the analysis of a wide variety of topics in environmental economics. We first pause to define what applied structural modeling is. Put simply, we consider it to be any empirical model where one recovers the parameters of behavioral functions—e.g., demand and supply equations, indirect utility functions of individuals, or the cost structures of firms. One typically begins with a theoretical model, being explicit about what can and cannot be observed by the researcher. Those elements of the model that cannot be observed are then classified as structural errors. These errors are often associated with unobserved heterogeneity across people or choice alternatives, and they are usually “backed-out” from observed behavior for a particular guess at the values of the behavioral parameters. Exactly how this is done will differ tremendously across applications. It could, for example, require simulating dynamic decision making or, alternatively, solving for the equilibrium of a strategic game. These errors allow for the estimation of behavioral parameters by way of an econometric procedure (e.g., generalized method of moments or maximum likelihood) that provides a criterion function on which to evaluate the quality of any particular vector of parameters.

2.1. The Advantages of Structural Modeling

In this section, we list a number of the key advantages associated with structural estimation techniques. Some brief references are made to papers that exploit these advantages, but detailed discussions of particular applications are left to Section 2.4.

Combining environmental systems with human behavior can lead to models with complicated feedback effects. Very often, environmental economists find themselves modeling equilibrium phenomena. Structural models are able to address explicitly those feedbacks. This can have a number of advantages. For example, many of the covariates in these models will be endogenous (e.g., congestion in a recreation demand or Tiebout sorting model). The structure of the model may suggest instrumental variables strategies to deal with those endogenous variables and recover consistent estimates, where otherwise valid instruments may be hard to find. See, for example, Murdock & Timmins (2007) and Bayer & Timmins (2007), which are discussed in further detail below.

Considering large policies (exactly what we mean by “large” will become clearer in the context of the applications we consider), welfare impacts may need to account explicitly for feedback effects. For example, a change in environmental quality in a neighborhood may induce people to re-sort across neighborhoods in a Tiebout fashion, changing important endogenous local attributes such as racial composition. Other neighborhood attributes (e.g., crime, school quality, traffic, and proximity to open space¹) may vary with the socioeconomic characteristics of residents as well. In a different context, fish stocks affect

¹See Walsh (2007) and Klaiber & Phaneuf (2009) for examples of models that evaluate open-space policies accounting for the endogeneity of private land conversion decisions in an equilibrium framework.

the allocation of fishing effort, but fishing effort affects those stocks through the harvest. An exogenous change in stocks or the cost of effort will set in motion the re-equilibration of the entire system.

Especially in the environmental context, feedback effects can work through natural/biological systems. Human behavior will impact biological processes (differentially in different places), and biological processes can induce varied, highly nonlinear feedback effects. Structural models allow one to make out-of-sample predictions of the impact of policy when feedback effects and nonlinearities are important. Of course, being able to make such predictions often relies crucially on assumptions one has to make about the functional form of behavioral equations, the distributions of unobservables, and other modeling assumptions. Looking for ways to test the validity of these assumptions (e.g., by investigating a model's ability to predict outcomes that are not used for model identification) is one of the most important areas for future research in structural modeling.

More generally, structural models can be useful when it comes to monetizing welfare effects, which is critical for doing cost-benefit analysis when the sources of benefits are diverse. For example, consider gasoline taxes versus CAFE (corporate average fuel economy) standards (a comparison we return to below). The former will operate mostly through the intensive margin (i.e., how often to drive), whereas the latter operates primarily through the extensive margin (i.e., which car to purchase or, more generally, whether to drive at all). Effects on each margin can have important policy implications. More generally, environmental economists need to combine health effects, amenity effects, effects on the quality of recreational uses, employment effects, etc., to value fully the welfare effects of a policy. Structural modeling allows us to convert many effects into a common denominator (i.e., dollars) through the lens of a utility function and an estimate of the marginal utility of income.

There are a number of other ways in which writing down a structural model can be useful in environmental contexts. First, doing so allows for the explicit modeling of heterogeneity in preferences and budget constraints. This can be important because heterogeneity can impact the effectiveness of a policy and its distributional impacts. Jacobsen (2008), discussed in further detail below, modeled the dynamic interactions between CAFE standards and the used-car market. Doing so, he found that it is not rich, new-car buyers who ultimately bear the costs of the policy, as is typically assumed, but rather the poor, who disproportionately buy used cars.

Second, in many environmental applications, policies affect interlinked decisions—e.g., whether to buy a fuel-efficient appliance or not (the extensive margin) and how much to use it (the intensive margin). Heterogeneity matters in this context as well—in particular, we do not want to treat these decisions as separate, because people who are more likely to buy a more efficient appliance may be disproportionately more or less likely to use it more often. The same structural parameters and sources of heterogeneity enter into both decisions. Bento et al. (2009) showed that this plays an important role in calculating the welfare effects of a gasoline tax.

The difference between the intensive and extensive margins is also important in the analysis of recreation demand (i.e., travel-cost models, which use the time costs and other expenditures incurred when visiting a recreation site to value the attributes that it provides). Typically, any particular recreator does not visit most sites—rather, visitors tend to make a few trips to a small subset of sites. Random-utility models are appropriate for studying decisions over a large number of differentiated alternatives, but they become

more difficult to apply when multiple decisions made by the same individual are not independent.² In contrast, traditional demand systems cannot handle the many zeros typical in these data. Instead, estimators have been developed that make explicit the use of the Kuhn-Tucker conditions that arise because consumption cannot be negative (Phaneuf et al. 2000).

Finally, structural models may often be able to predict or explain unintended consequences of a policy. The strength of a structural model is its ability to simulate counterfactual scenarios. One is able to “turn off” a piece of the policy environment to see how the world would have played out in its absence. This can be important when the policy interacts in a complicated way with the rest of the economic system (e.g., a pollution-control policy that also functions as a barrier to entry that restricts competition).

2.2. Structural Modeling and Tiebout Sorting

The structural approach is particularly useful in the class of models that rely on Tiebout’s idea that, by “voting with their feet,” individuals reveal their preferences for local public goods and bads. Originally, such models fell into the category of hedonic analysis, a two-stage procedure that first applies reduced-form empirical methods to estimate a hedonic price function. That hedonic price function is then combined with utility theory, which is used to recover preferences in the form of a marginal willingness-to-pay (WTP) function (Rosen 1974). Although the hedonic technique is widely used in academic, government, and consulting applications, it exhibits several well-known problems, many of which can be addressed by applying structural techniques. We now discuss nine instances where we believe this to be the case.

2.2.1. Omitted variables bias. If important unobserved local attributes covary with the local attribute of interest, the researcher estimating the first-stage hedonic price function will recover a biased estimate of the marginal value of that attribute (Small 1975). One of the simplest solutions to this problem is to exploit variation in that attribute over time, assuming other attributes remain fixed (i.e., house or neighborhood fixed effects). The reduced-form literature has produced a variety of other approaches to deal with the omitted variables problem (see, for example, Chay & Greenstone 2005), whereas instrumental variables solutions have also been incorporated into structural models (e.g., Bayer et al. 2009).

2.2.2. Identification. The simple linear-quadratic hedonic model does not identify the marginal WTP function (Brown & Rosen 1982). Solutions to this problem have been found in the use of data from multiple markets or functional form restrictions (Mendelsohn 1985), although more recently Ekeland et al. (2004) showed that the linear-quadratic model is a very special case and that, by being flexible about the first stage, the identification problem goes away. Without careful structural modeling of the individual utility maximization decision (checking to see if second-order conditions are satisfied), however, the flexible first stage can yield misleading estimates (Bishop & Timmins 2008).

²For example, one may redefine combinations of different choices to be new elements of an expanded choice set. If individuals make relatively few decisions [e.g., where to work and where to live, as in Kuminoff (2008)], this will be feasible. For individuals choosing how to allocate a large number of potential recreation opportunities over many different sites, the size of the redefined choice set will become intractable.

2.2.3. Endogeneity. The individual simultaneously picks both the quantity of an amenity and its hedonic price (assuming that price is allowed to vary with the quantity consumed), meaning that the simple hedonic estimation of the marginal WTP function will yield biased estimates (Bartik 1987, Epple 1987). Valid candidates for instrumental variables can be hard to find. Bishop & Timmins (2008) showed how this problem can be avoided with a structural model of the choice of the attribute subject to a nonlinear budget constraint.

2.2.4. Endogenous attributes. Many attributes of interest are formed by the equilibrium process of individuals sorting in a Tiebout fashion (e.g., crime rates, pollution caused by traffic flows, emissions from production activity). As alluded to above, structural modeling may suggest instrumental variables strategies for many of these attributes. Specific examples are described below.

2.2.5. Re-equilibration. In response to large changes in amenities, people may reoptimize. To model welfare effects, we need to know how they do that. Simply modeling the equilibrium hedonic price function is insufficient. This has been the main area where structural models of Tiebout sorting have proven useful. See, for example, Sieg et al. 2004 and Murdock & Timmins 2007.

2.2.6. Choice set. The hedonic model assumes individuals are free to maximize utility with respect to every attribute that enters into utility (i.e., the choice set is dense). In reality, the choice set may contain “holes”—certain combinations of attributes may be unavailable at any price, and individuals are forced to choose the best option from the discrete set of available opportunities. In contrast to the hedonic framework, structural models of Tiebout sorting typically begin from the premise that individuals maximize utility from a discrete set of available choices (e.g., houses or jurisdictions).

In addition, sorting models give the researcher the ability to model interrelated choice sets (e.g., where to live, how much housing to purchase in that location, and where to work) in a straightforward way (Kuminoff 2008). The decisions made in one of these dimensions can have important implications for how responsive individuals will be to policies designed to influence their behavior in other dimensions.

2.2.7. Dynamics. Hedonic models do not treat individuals as though they are forward-looking. However, given the expense incurred and effort involved in buying a house, we may expect that people will care about more than just the current attributes of the house and its neighborhood—they may base decisions on expectations about how those attributes will evolve over time. In the classic dynamic estimation framework developed by Rust (1987), this would involve a computationally prohibitive state space. Recently developed two-step estimators (Hotz & Miller 1993, Arcidiacono & Miller 2008) have helped to avoid these problems. Bishop (2008) applied these new methods to the sort of intercity sorting problem described by Roback (1982). Bayer et al. (2008b) adapted them for use in describing intracity sorting across houses.

2.2.8. Stickiness. Hedonic models do not generally allow for “stickiness” in individual decision making (e.g., transactions costs in a house-price model or moving costs in a wage-hedonic framework). We logically expect that people relocate only when the utility gain

from doing so exceeds those transactions costs. In the wage-hedonic context, research shows that people tend to live close to the region of their birth (Bayer et al. 2009), creating a tension within hedonic theory. In particular, the model assumes that individuals face no impediments to mobility in the formation of the hedonic equilibrium, while assuming that no one would reoptimize in response to a large policy. Sorting models, which model the processes that yield the hedonic equilibrium rather than the equilibrium itself, allow for stickiness to be incorporated explicitly into the hedonic equilibrium. For example, Bayer et al. (2009) modeled long-run migration costs (i.e., location at ages 25–35 relative to birth location), whereas Bishop (2008) also included short-run migration costs (i.e., current location relative to location two years prior) in her modeling of forward-looking agents. Moreover, she included search costs (another form of stickiness) in the manner of Kennan & Walker (2005), by allowing people to have better information about labor markets in the areas where they had lived in recently.

2.2.9. Endogenous outcomes. When sorting over labor markets, as is the case in the wage-hedonic model described by Roback (1982), observed wage distributions provide a distorted representation of underlying (unconditional) wage distributions. This insight is from the Roy (1951) model, where individuals sort across employment sectors based on common returns (i.e., returns to education or experience) and idiosyncratic returns (i.e., individuals tend to go where their individual-specific returns are high, shifting up the conditional wage distribution). The complicating factor is that individuals also sort based on nonwage factors (i.e., amenities). Bayer et al. (2008a) described the problems that arise in identifying the role of these amenities and showed how they (and the wage distributions from which sorting individuals receive draws) can be recovered nonparametrically. Deleire & Timmins (2008) employed a similar approach to model the sorting of workers across occupations that underlies the wage-hedonic measurement of the value of a statistical life, accounting for Roy sorting. This is akin to the classic Roy job-sorting application; the main difference is that people are allowed to care about nonwage job attributes such as fatality risk. Controlling for the biases caused by Roy sorting raises the value of a statistical life estimated by Deleire & Timmins (2008) by as much as a factor of 4, compared with the traditional wage-hedonic estimate.

2.3. Drawbacks of Structural Approaches in Sorting Models

It is worth pausing at this point to mention some of the drawbacks of structural approaches to modeling Tiebout sorting behavior. While able to deal with many of the problems described above, a tractable sorting model requires the researcher to specify a choice set, define limits on mobility, establish parametric restrictions on utility, and impose distributional assumptions on random parameters. These modeling choices are not inconsequential and should be verified whenever possible (e.g., by estimating the model with a subset of the data and checking its ability to predict behavior in the remaining data). Kuminoff (2009) investigated the role played by each of these structural assumptions in identifying nonmarket values.

Structural models used to predict the equilibrium impacts of large policy changes may require even more modeling assumptions. For example, a model of sorting over both labor and housing markets may require additional assumptions such as having the spatial distribution of employers and the supply of heterogeneous housing remain fixed in order to

calculate counterfactual labor and housing market equilibria. In the long run, these may be particularly poor assumptions. Although nothing, in principle, prohibits one from relaxing these assumptions, data and computational burdens may make doing so impractical.

2.4. Examples of Structural Estimation in Environmental and Resource Economics

Smith & Wilen (2003) re-examined the effectiveness of marine reserves—spatially delineated closures of fishing areas designed to prevent overexploitation and allow for regrowth of a fish stock. For years, reserves have been thought of as one of the most promising options for managing declining fisheries (as opposed to regulating the length of the fishing season or imposing gear restrictions). This impression, however, has been based largely on models that did not realistically describe how fishermen would respond to the imposition of reserve restrictions. Prior to this paper, models tended to assume that fishermen would reallocate effort evenly over the remaining unregulated fishing grounds. Working with data from the Northern California sea urchin fishery, Smith & Wilen instead used a behavioral model to describe how fishermen would reallocate effort; they then connected that model to a realistic description of biological processes.³ Factors affecting the decision of whether to fish (and if so, in which patches) include abundance, weather risk, price, travel distance, and the spatial distribution of closures. Both biology and behavior are important to policy conclusions. In particular, in the behavioral model, unattractive patches serve as *de facto* reserves. In combination with a declining overall effort on the part of fishermen in response to the imposition of a reserve, this leads to an increase in the production of sea urchin eggs. However, whereas the model that naïvely assumes a uniform redistribution of effort in response to a closure predicts an overall increase in harvest, the model that assumes an optimizing reallocation of effort predicts a reduction in total harvest, owing to the reduction in the optimizing level of effort.

Smith & Wilen (2003) provided an example of the importance of marrying behavioral and biological modeling, particularly when making out-of-sample predictions. Simply using the effects of an observed closure on harvests and egg production to predict the effects of a hypothetical closure may not work. Closing a patch that is particularly important for larval growth will have big impacts on stock and harvest. However, closing a patch in which fishermen have little interest will have little impact. The geography of closures matters as well. Because Northern California sea urchin larvae flow from north to south, closing a northern patch will have very different impacts on behavior and harvest outcomes than would closing a southern patch. Alternatively, a closure that reallocates effort to a patch that is important for larval production can be extremely damaging.

Finally, Smith & Wilen (2003) illustrated the importance of controlling for feedback effects in simulating equilibrium outcomes. Those equilibrium effects operate through biological channels. In particular, increased stock abundance from a closure may lead to increased fishing effort on that patch, thereby diminishing stock in the long run and reducing equilibrium effort. In equilibrium, the reallocation of effort will dampen the initial stock effect.

³From a biological perspective, sea urchins provide an ideal species with which to study these effects, both because they are “patchy” (i.e., sea urchins live in small, spatially delimited groups) and because there are important larval transmission mechanisms that impact growth rates. For both these reasons, spatial effects matter for sea urchins.

Murdock & Timmins (2007) looked instead at the behavior of recreational anglers, with the goal of determining the welfare effects of a “large” policy. This paper follows the established methodological approach of using the trade-offs made by individuals when choosing a recreation site to value site attributes. Along with expected catch, an important (but problematic) site attribute that may affect the recreation decision is congestion, i.e., the number of other anglers one expects to encounter. Congestion is problematic because it is likely to be correlated with site attributes that the researcher cannot observe. Crowded sites are likely to be crowded for a reason—they have attractive features that draw in anglers. The fact that anglers choose sites with attractive unobservables therefore means that they will also tend to choose congested locations. A naïve empirical model concludes from this that they derive utility from congestion.

This conclusion has important implications for valuing the welfare impacts of a large policy change. Murdock & Timmins (2007) considered the effects of closing Lake Winnebago to fishing. Closures (or at least fish consumption advisories) of this sort are typically caused by pollution, e.g., excessive contamination of mercury or polychlorinated biphenyls. Lake Winnebago is the largest lake located entirely inside Wisconsin (measuring approximately 30 by 10 miles in area). It is a popular fishing destination, known particularly for walleye. A closure of this magnitude would send a large number of anglers to substitute sites. A model that understates the congestion costs (or even incorrectly measures the sign of their welfare effect) would not properly account for this impact and would understate the cost of the closure. Properly recovering the unbiased contribution of congestion, Murdock & Timmins (2007) found that the welfare costs of the closure are approximately doubled.

In intuitive terms, congestion externalities can be viewed as a standard endogeneity problem. Even though the angler’s decision-making process is described with a random-utility model discrete-choice framework, the solution to the problem can still be expressed in an instrumental variables framework (Berry et al. 1995). The logic of instrumental variables estimation requires a variable that is correlated with the number of other visitors attending a particular site, but uncorrelated with the unobserved attributes of that site. This is where structural modeling enters the picture. With a guess at the parameters of the angler’s random-utility function, it is possible to build a prediction of the number of fishermen who would choose each site on the basis of only observable, exogenous attributes (i.e., ignoring congestion and unobservables). By construction, that prediction will be uncorrelated with the unobservable attributes of the location in question, but (assuming observable, exogenous site attributes are consequential in the angler’s decision-making process) it will be correlated with the congestion variable. Bayer & Timmins (2007) described this strategy for constructing instruments (which is useful in the cases of congestion, agglomeration, and endogenous neighborhood attributes based on socioeconomic factors and demographics) in more detail.

There are other empirical strategies used in the environmental economics literature to deal with the problem of measuring the spillover impacts of congestion and agglomeration. Hicks et al. (2008) considered the case of commercial fishermen in the Bering Sea, where the effect of congestion on profits may be nonmonotonic. They employed an instrumental variables strategy similar to that used by Murdock & Timmins (2007), but they dealt with a far more complex definition of equilibrium. Phaneuf et al. (2009) demonstrated how the problem can be addressed in the context of the Kuhn-Tucker estimation models described above. Boxall et al. (2005) dealt with a similar problem

(modeling congestion in Canadian wilderness canoeing), but they employed a very different remedy for the endogeneity problem. Whereas the approach described above relies on a great deal of choice-set variation to construct powerful instruments, Boxall et al. (2005) showed how an exogenous source of variation can be obtained by combining stated-preference and revealed-preference data. The result is an approach to measuring congestion that is particularly valuable when the choice set is small. Also dealing with a small choice set, von Haefen & Phaneuf (2008) demonstrated how stated-preference data can be combined directly with the empirical strategy described in Murdock & Timmins (2007) to achieve identification. Finally, Pfaff & Robalino (2005) modeled spillover effects in neighbors' decisions about land clearing in Costa Rica. This is another spatially localized form of agglomeration effect. They employed land slope as an instrument, i.e., an exogenous determinant of a neighbor's deforestation decision that can only affect one's deforestation decision through the neighbor's decision. Doing so, they found important evidence of spatial spillovers. This has practical importance for policy makers, as policies designed to reduce deforestation may reduce deforestation activities on plots not directly targeted by those policies.

In no other area of environmental economics are feedback effects more important than when sorting behavior in housing or labor markets is used to value nonmarketed amenities. This is the basis for the hedonic approach, which uses the information in the outcome of the sorting process (i.e., the equilibrium hedonic price function) to value goods ranging from processor speed in computers to human lives, and amenities ranging from air quality to Superfund site proximity. Given the well-known problems with the hedonic method cited in Section 2.3, a new class of estimable Tiebout sorting models has evolved. These models contrast with the traditional hedonic models in that they focus on the process that leads to equilibrium, rather than relying only on the information in the hedonic equilibrium. As a result, many of the problems associated with the traditional hedonic model may be addressed directly.

The first models of local public goods provision and residential location, based on the original ideas of Tiebout (1956), were developed by Epple et al. (1984), Epple & Romer (1991), Fernandez & Rogerson (1996), and Nechyba (1997a, 1997b). An estimation framework for this class of models was then developed by Epple & Sieg (1999). Sieg et al. (2004) used that approach to value air quality in Southern California, paying particular attention to the fact that, were air quality to change nonmarginally, people may find it optimal to live elsewhere (with subsequent changes in the equilibrium prices that clear the housing market). The intuition for the Epple-Sieg estimation strategy is to build a full equilibrium model of the location decisions of heterogeneous households (who differ by income and preferences for public goods), requiring housing prices to adjust to clear the market in every jurisdiction. Two key features of the model facilitate its use. First, all households are assumed to place the same utility weights on different local public goods. For example, no matter whether one has children or not, one puts the same relative weights on air quality and school quality. Everyone is therefore able to agree on a quality ranking for jurisdictions, and all can then make decisions based on how they value jurisdictional quality relative to private consumption. Because of the similarity of this assumption to vertical models of differentiated product demand in industrial organization, this has come to be known as the class of vertical sorting models.

Second, conditional upon having the same preference for jurisdictional quality relative to private consumption, individuals will stratify according to income (i.e., high-income individuals will locate in higher quality jurisdictions). This is a result of a single-crossing

property exhibited by utility.⁴ The model is closed in the sense that, given a vector of structural parameters, it can predict population, income distributions, housing consumption, and (subject to a normalization) the index of jurisdictional quality in every location. Estimation proceeds by matching elements from the distributions of the first three to observed data. The index of jurisdictional quality can be further decomposed to determine the value individuals place on particular local attributes (e.g., air quality).

In their application, Sieg et al. (2004) considered 103 school districts located in five Southern California counties, and they used data describing housing transactions between 1989 and 1991. They interpolated air quality measures for individual houses, using monitor data for ozone and particulates, and they considered school quality measured by the average math score on the 1992–1993 California Learning Assessment System Grade Level Performance Assessment Test. After recovering structural utility parameters, they used their model to value the welfare improvement associated with the large changes in ozone concentrations in the area between 1990 and 1995 (i.e., a 13.9% average reduction) in both a partial equilibrium (PE) setting (i.e., assuming people do not re-equilibrate) and in a general equilibrium (GE) context (i.e., allowing people to reoptimize in their location choices). In calculating the GE welfare gain, the paper asks the following question: After re-equilibration, how much compensation would everyone need to be given (or have taken away) in order to be returned to her pre-policy level of utility? Calculation of this utility theoretic measure of welfare relies heavily on having recovered estimates of structural utility parameters. It also requires the strong assumption that no individuals are allowed to enter or leave the five counties being considered—i.e., only the decisions of those present at the start of the decade are modeled.

While the average welfare gain in the GE context is only slightly higher than that in the PE context (i.e., \$1371 versus \$1210), the difference between the two measures can be very large for certain individuals. For example, those in the least expensive school district, who (conditional on income) have revealed a preference for private (relative to public) goods consumption, will have nowhere else to go as the price of housing starts to rise because of improved air quality in their district relative to that in other districts. Assuming these individuals are renters, they are made worse off in the GE model relative to the PE model. Alternatively, those starting in affluent locations with good public goods will do better in the GE framework, as the presence of more quality substitute locations lowers the price they must pay for a home.

Based again on comparisons with the literature on differentiated product demand in industrial organization, the alternative approach to modeling Tiebout sorting behavior is classified as the horizontal approach. Whereas the vertical model required all individuals to agree on a quality index associated with each jurisdiction, the horizontal model allows each individual to potentially place a unique value on each component of the jurisdictional public good. In most applications, individuals are also given an idiosyncratic preference for each jurisdiction. If this preference is drawn from a Type II extreme value distribution, the horizontal location choice model can be written as a logit. If all individuals are allowed to have preferences for each component public good drawn from a random distribution, this is a random parameters logit. If preferences vary only by observable individual attributes, it is described as a heterogeneous parameters logit.

⁴In technical terms, single crossing requires the slope of indifference curves in amenity-housing price space to vary monotonically with income, conditional on preferences (and vice versa).

What makes the horizontal sorting model different from a simple multinomial logit model of location choice are the complications imposed by equilibrium. In particular, many of the attributes that are important for location choice are determined in equilibrium by the location-choice process (e.g., neighborhood racial composition, housing services prices, agglomeration or congestion effects in intercity sorting). If these variables are important factors in location choice, then properly estimating how they enter the utility function will be crucial to making accurate GE predictions of the welfare effects of a large policy change. An identification strategy similar to that described in Murdock & Timmins (2007) for recovering the effect of congestion on recreational demand can be employed.

Bayer et al. (2009) used a simpler version of the horizontal model (i.e., one that considers only marginal changes) in a paper that focuses on the role of migration costs in valuing air quality. A casual inspection of Census data describing long-term migration behavior shows that most U.S. residents do not move far from where they were born. For example, between 58% and 79% of household heads were found to be living in the Census division of their birth in the 2000 Census.⁵ However, if individuals tend to be born in locations with relatively poor air quality (e.g., the Northeast and Rust Belt), then their failure to move to relatively clean states in the West would bias downward estimates of the disutility associated with pollution, unless these migration costs are modeled explicitly. This paper includes explicit migration costs in the sorting process; doing so nearly doubles the estimate of marginal WTP to avoid a small increase in particulate matter.

Neither the vertical nor the horizontal approach to modeling equilibrium sorting is ideal. On the one hand, the pure vertical model requires all individuals to agree on a relative ranking of jurisdictions based on factors besides housing prices, which is not realistic. Different people will have varying work locations and commuting costs can be important, not to mention the fact that different types of people (e.g., with and without children) will care differentially about public goods (e.g., school quality). On the other hand, horizontal models, which handle these sorts of preference heterogeneity very well, rely to a greater extent on distributional assumptions for identification. In particular, in a vertical model, only neighboring jurisdictions in quality space are potential substitutes, whereas in a horizontal model, every location is one. This feature arises because the horizontal model includes an idiosyncratic preference shock. Because of this shock, every individual has a positive probability of choosing every location.

In the process of building a hybrid model, Kuminoff (2009) explained these trade-offs in clear terms. His Epple-Sieg equilibrium framework nonparametrically bounds individual utility parameters (Epple et al. 2004). By parameterizing the distribution of preference heterogeneity and income, these bounds imply a likelihood function that could be maximized for parameter point identification. Different people can therefore have different relative preferences for the same public goods. Moreover, in common with many horizontal models, Kuminoff (2008) explicitly modeled the fact that individuals choosing over a large geographic area will choose over both housing and labor markets (i.e., where one chooses to live impacts the labor market opportunities available to that individual). In an application to sorting over school districts in Sacramento, California, and the San Francisco Bay Area (close enough that they may be considered to be in the same residential choice set, but far enough

⁵The Census Bureau divides the United States into nine divisions: New England, Mid-Atlantic, East North Central, West North Central, South Atlantic, East South Central, West South Central, Mountain, and Pacific.

apart that they constitute separate labor markets), Kuminoff found that incorporating the labor market decision significantly expands the upper bound on possible marginal WTPs for ozone. Many people living in places with high levels of ozone do so because it affords them access to better labor market opportunities. Moreover, by adding preference heterogeneity to the vertical model, the estimated bounds on preference parameters are widened. If one were to use these bounds to carry out maximum likelihood estimation, this would imply that the resulting point estimates would become more sensitive to the assumed distribution of individual preference heterogeneity.

Jacobsen (2008) is an outstanding example of how structural modeling can be used to (a) jointly model multiple interlinked decision processes, (b) explicitly control for the role of heterogeneity in decision making and determine how that heterogeneity affects policy outcomes and incidence, and (c) combine the many complicated effects of policy to arrive at a single welfare estimate that can be used for cost-benefit analysis. Jacobsen (2008) explored the relative effectiveness of CAFE standards and the gasoline tax as tools for reducing gasoline consumption (for an application to digital cameras, which are differentiated only by pixels, see Carranza 2008). Properly modeling these policies requires a model of price setting by imperfectly competitive firms that are differentially affected by tightening CAFE standards. That model must also describe consumers' decisions about which car to buy and how much to drive it. Finally, it needs to account for interactions between the new- and used-car markets.

Modeling the car-purchase and driving decisions is especially important for this application, because most of the benefits of the CAFE standards come on the extensive margin, whereas the gasoline tax matters primarily on the intensive margin. Considering these decisions together is important because of the rebound effect (i.e., after buying a more fuel-efficient car, made less expensive by stricter CAFE standards, the individual may actually choose to drive it more than she would have driven a less fuel-efficient vehicle, thereby reducing the fuel savings).

Incorporating the used-car market turns out to be particularly important, for evaluating both the effectiveness of the CAFE standard and its incidence. When buying a car, consumers have the option of going to either the new- or used-car market. Because of income effects, richer individuals are more likely to purchase a new vehicle. Because the immediate effect of the CAFE standard is to raise new-car prices, the conventional wisdom is that the burden falls mainly on the rich (i.e., a progressive policy). However, modeling the interaction of the new- and used-car markets over time paints a very different picture. Increased CAFE standards today reduce the flow of larger cars to the used market, raising the price of those vehicles and making it more costly for a used-car consumer to buy one. Moreover, the used-car fleet becomes more inefficient as large used cars become more scarce and are less likely to be scrapped.

Finally, looking at the effect of strategic pricing decisions between imperfectly competitive firms, many of the gains of tightened CAFE standards are undone by the fact that violating types (i.e., European importers) find it optimal to sell more large cars to fill the void in the market left by the constrained domestic producers who raise the miles per gallon of their fleet in response to the policy. These firms enjoy increased profits at the expense of the U.S. producers, who bear most of the costs. Accounting for these market equilibrium effects, Jacobsen (2008) found that CAFE standards do a relatively poor job of reducing fuel consumption. Combining all of these effects, the gasoline tax can achieve the same reduction in gasoline use at one-sixth the cost.

2.5. Using Structural Models to Measure the Unintended Consequences of Environmental Policy

We now turn to two papers that use structural models of dynamic decision making to illustrate the unintended consequences of an environmental policy. Very often, dynamic incentives lead to outcomes and policy incidence that differ greatly from that which would be expected on the basis of a simple static analysis. In our first example, Ryan (2006) used a structural model of dynamic decision making to simulate the world in the absence of air quality regulation. He measured the dynamic impacts of the 1990 Clean Air Act Amendments (CAAA) on the producers of Portland cement. Making Portland cement is an extremely polluting activity, and the 1990 CAAA increased the certification requirements for new producers. Although the policy resulted in no immediate changes in variable production costs, the regulation was expected, in the long run, to reduce emissions from the industry. The policy also had the effects, however, of increasing entry costs and reducing entry. These outcomes enabled incumbents to raise prices on consumers.

To measure these unintended consequences, Ryan (2006) required a model of the process governing entry into the industry. Assuming the industry is made up of many small markets, he modeled the strategic pricing, investment, and exit decisions of incumbents, along with the entry decision of a potential new entrant. This is a difficult task because of the dynamic nature of the game played between firms. Typical of the industrial organization literature on dynamic games, Ryan assumed that the price-setting decision is made in a static game played conditional upon dynamic capacity investment decisions. Investment decisions are, however, made given expectations about the investment decisions of rivals. Investment determines capacity, which, in turn, determines production costs. The capital stocks of each firm (both incumbents and potential entrants) are state variables, and solving the game with traditional methods (e.g., Pakes & McGuire 1994) requires the repeated solution to a very high-dimension dynamic programming problem. Because solving this problem even once takes a very long time, embedding it in an estimation algorithm is impractical (i.e., solving it repeatedly for different parameter guesses). Instead, Ryan (2006) employed the estimation algorithm suggested by Bajari et al. (2007) to avoid this problem.

Part of a new wave of two-step estimators (see also Pesendorfer & Schmidt-Dengler 2008, Aguirregabiria & Mira 2007, Pakes et al. 2004), Bajari et al. (2007) recovered the structural parameters governing firms' investment decisions without ever actually solving the dynamic game. The intuition is as follows: First, the researcher recovers simple reduced-form estimates of the firm's policy functions (e.g., how much should the firm invest and when should it exit, conditional upon any set of market circumstances) directly from the data, being careful to impose as little structure on those estimates as possible. The firm's behavioral structural parameters can then be estimated by guessing at values for those parameters and then simulating behavior according to the policy functions. Assuming that the policy functions were indeed optimal, this simulated behavior should yield a larger net present discounted value than would any other policy function (e.g., one defined by making a slight perturbation to the observed policy function). The researcher searches over structural parameters until the observed policy function tends always to generate higher discounted net present value payoffs than any other perturbed policy function. Because the game is never solved (only simulated on the basis of reduced-form policy functions), the "curse of dimensionality" associated with the large state space does not

come into play. Ryan (2006) found that the sunk cost of entry rises because of the new certification requirements imposed by the CAAA from \$46 million to \$58 million.

Having estimated the parameters governing this dynamic game, Ryan (2006) simulated how the industry would have evolved in the absence of the CAAAs in order to determine how strategic play (in particular, entry and subsequent pricing decisions) would be altered. Specifically, he simulated entry and pricing behavior in a typical market after the CAAAs and then did so again while assuming that the post-CAAA sunk entry cost had remained equal to its pre-CAAA level. As a result of the increased entry costs, prices rise by 6%, whereas quantities fall by 16%. With a 90% drop in entry, consumers take a welfare penalty of approximately \$29 million, whereas producers benefit from the reduced competition by just under \$3 million. Although the impression had been that the CAAAs were a burdensome regulation for all firms, this model of strategic entry and investment behavior suggests otherwise. The barrier to entry imposed by these regulations proved to be a net benefit for incumbent firms, who face less competition in the price-setting game.

Note that the two-step approach to estimating the structural parameters of a dynamic game does have its limitations. In particular, to simulate a counterfactual scenario (e.g., entry in a world in which entry costs had not risen), the solution to the dynamic game does need to be solved (at least once). The size of the state space then becomes an issue. In practice, this limits the number of firms (both incumbents and potential entrants) that can be considered in the simulation to four or five. Beyond this point, the curse of dimensionality caused by the expanding state space makes the problem computationally infeasible. This limit allows for fewer firms in a single market than is typical in many industries. Benkard et al. (2007) have proposed a solution to this problem in which firms ignore the identities of the firms they are playing against.

We conclude this section with a brief discussion of a second paper that uses dynamic estimation techniques to recover the unintended consequences of clean air regulations. Heutel (2008) estimated a model of the decision by power-plant owners to scrap and rebuild as either a coal or noncoal plant, to install a scrubber, or to switch to low-sulfur coal in response to the 1990 CAAAs. Plants built prior to the amendments, however, had been grandfathered (i.e., made exempt) from the rules contained in the New Source Pollution Standards. Scrapping and building a new generator would void their grandfathered status, causing forward-looking plant owners to postpone that decision beyond the time when it would normally be optimal. By recovering the parameters governing these dynamic decision processes, Heutel (2008) simulated how investment behavior would have changed had grandfathered status never been granted: He found that SO₂ emissions would have fallen by 60% by 1995.

3. REDUCED-FORM APPROACHES

Although the term reduced form is used frequently, the underlying definition has evolved over time. The traditional econometrics textbook definition of reduced form refers to simultaneous equations, where the system of equations is solved to eliminate all endogenous variables. The reduced-form model links the dependent variable solely to exogenous variables, which, by definition, are not influenced through feedback loops of the system. For example, both demand and supply equations depend on price as well as on other exogenous demand/supply shifters. The reduced-form model eliminates the price variable and derives the equilibrium quantity solely as a function of the demand/supply shifters,

which presumably are exogenous. The emphasis is on eliminating endogenous variables, whereas a structural setup would model and estimate the effect of such variables directly. Modeling such endogenous variables usually requires structural assumptions to identify these variables of interest as well as suitable instruments. The goal of a reduced-form model is to avoid as many structural assumptions as possible. It should be noted that almost all models rely on some assumptions, and reduced-form models require assumptions of their own, e.g., additive separability and conditional independence in dynamic discrete choice settings (Rust 2008).

Using the traditional econometrics textbook definition, a cross-sectional study that relates a variable of interest to exogenous characteristics that vary in space would be considered reduced form. For example, a hedonic model that, following Rosen (1974), estimates the marginal WTP for a public good by linking house prices to different levels of the public good across space at a given point in time would fall under such a classification.

More recently, the definition of reduced form has sometimes been narrowed further and become synonymous with program evaluation or treatment effects, where further emphasis is placed on the identification of the exogenous variables. The researcher is interested not only in limiting the model to exogenous variables, but also in how these variables are assigned. The primary concern is that exogenous variables may be correlated with other confounding variables, and hence the estimated coefficient on the exogenous variable would be biased as it picks up both the direct effect of the exogenous variable and the effect of the collinear variables that are omitted from the analysis. This narrower definition is a natural extension of the aforementioned goal to make the results as independent as possible of structural assumptions. A pure cross-sectional approach requires that all confounding factors be correctly accounted for to derive an unbiased estimate.

Various research designs are used to avoid a potential correlation with the omitted variables. A nonexhaustive list includes the following:

1. A panel setting that can allow a researcher to use fixed effects to capture all time-invariant additive factors, thereby accounting for some baseline differences.
2. Quasi-experiments that are conducted under conditions where the exogenous variables are arguably randomly assigned and orthogonal to other variables of interests. For example, weather is random and not influenced by economic agents: “Good” and “bad” weather outcomes over time should be orthogonal to other controls.
3. In a regression discontinuity framework, a treatment variable gets “switched on” once a threshold is passed. Observations just below the threshold are compared with observations right above the threshold. The important underlying assumption is that there is no stratification at the threshold, and hence observations right below it are similar to the ones right above it, except for the triggered treatment.
4. Randomization, which assigns a treatment deliberately to a subsample of the entire population, while the remaining observations that were not assigned the treatment act as a control group. The randomization of the treatment ensures that it is uncorrelated with other variables and that the coefficient on the treatment variable will be unbiased.

3.1 The Advantages of Reduced-Form Modeling

Before we start our overview of reduced-form studies, it is helpful to remind the reader what can and cannot be accomplished with regression analysis. Freedman (1991) discussed the use of regression models in the social sciences by elaborating the virtues and

vices of both structural work and reduced form models. He argued that “causal arguments based on significance test and regression are almost necessarily circular. To derive a regression model we need an elaborate theory that specifies the variables in the system, their causal interconnections, the functional form of the relationships, and the statistical properties of the error terms—independence, exogeneity, etc.” (Freedman 1991, p. 292). Such a line of reasoning has more in common with the structural setup, which starts with a broad model that relies only on primitive variables and is guided by theory on the functional form. Freedman then continued, “There is an alternative validation strategy, which is less dependent on prior theory: Take the model as a black box and test it against empirical reality. Does the model predict new phenomena? Does it predict the results of interventions? Are the predictions right?” (p. 293). Such a strategy is more in line with a reduced-form approach. Are reliable predictions feasible with as few assumptions and structural feedback mechanisms as possible? Common themes in reduced-form studies hence are as follows.

3.1.1 Identification. Economics shares the problem of most other social sciences, which all rely predominantly on observational data. For that reason, great care has to be exercised in identifying the parameters of interest to ensure that the regression is not picking up some spurious correlation. A clever identification strategy that establishes some exogenous variation can be more convincing than a model that aims at incorporating all possible confounding variables explicitly. It is difficult to include all possible confounding variables in a structural model and free it from potential omitted variables bias.

3.1.2. Transparency. Reduced-form models make it easier to understand what source of variation explains the source of variation in the dependent variable. They are often simpler to estimate from a statistical perspective, and simplicity is a virtue. It is more straightforward for the reader to see which variables are ultimately responsible for the new outcome.

3.1.3. Predictive power. Examples from macroeconomics literature show that complex equilibrium models perform worse than simple univariate time-series models when predicting future economic outcomes (e.g., Wallis 1989, Granger & Yongil 2003). Errors can feed through structural models and propagate. Both reduced-form and structural models should be judged by how well they can predict empirical phenomena. One important qualification of the goodness-of-fit measures is in order: Generally, testing out-of sample performance is superior to in-sample measures like R-square. The latter can “over-fit” the data. To give an illustrative example of a model with a great in-sample fit that does not extend out of sample, consider the following: Before 2006, Germany won the soccer World Cup if and only if the final was played on a single-digit July date. The R-square of this model is 1 as it gives a perfect prediction of Germany’s success. In 2006, Germany hosted the World Cup, and in accordance with this model, the final was played on a single-digit July day. However, although the in-sample prediction was perfect, it did not extend out of sample, as Italy won the World Cup. Although this is an anecdotal story, it illustrates the point that models that can be fit to a set of data and give very good fits do not necessarily provide equally convincing out-of-sample forecasts.

3.1.4. Robustness. Highly nonlinear structural problems require numerical solution techniques to approximate extrema, and it is not always evident that the chosen extreme was indeed the global optimum (Knittel & Metaxoglou 2008).

Freedman (1991) cited the work of Snow (1965) as a prime example of a data-driven reduced-form approach. Snow discovered that cholera is a waterborne disease. Although Snow did not run advanced regressions, he did a lot of “legwork.” He realized that one apartment building whose water supply was contaminated with runoff from privies had many more cases than an adjacent building with a clear source. This was a cross-sectional analysis. Subsequently, he discovered that three water companies serviced London. One company received water from the Thames above the major discharge areas, whereas the other two obtained water from sources below the discharge areas. The type of constituents serviced by each company were indistinguishable, yet the cholera rate among the former was much lower than among the latter. In a sense, Snow discovered a quasi-experiment: Because the water agencies served customers throughout London, the subsets were balanced (same type of households, incomes, etc.), and the only difference was the water source. Difference in cholera rates between the different samples could convincingly be linked to the water source. Snow exercised great effort to come up with a plausible identification strategy. Once he had established the hypothesis that cholera was a waterborne disease, he tested it with various data sets to confirm it. Different reduced-form approaches are discussed in further detail in Section 3.2 below.

Chetty (2008) recently contrasted structural and reduced-form approaches and discussed how they can inform one another. Structural models are often needed for welfare analysis, yet some analyses can be answered with the help of a few high-level parameters, i.e., these parameters are sufficient for the analysis. For example, a Harberger deadweight loss triangle of taxation depends solely on the elasticity of demand for a good with respect to a tax, and hence that elasticity is all that is required for a study of the welfare impacts of taxation. There may be various combinations of primitives that all give the same elasticity, but the primitives have no impact on the ultimate welfare estimate. Whenever a welfare analysis can be reduced to a set of sufficient high-level statistics and it does not require the researcher to identify the underlying primitives of the structural model, reduced-form studies can be especially valuable in identifying them with as few assumptions as possible and a high degree of transparency. An example of a key parameter best suited for reduced-form studies is the effect of any exogenously changing environmental variable on an economic or natural indicator, e.g., the impact of exogenous pollution variation on health, housing values, etc.

3.2. Examples of Reduced-Form Models in Environmental and Resource Economics

The following examples contrast various reduced-form approaches in more depth and briefly discuss the advantages and disadvantages of each approach in identifying key parameters. Readers interested in the latest developments in econometric techniques using reduced-form estimators are referred to Imbens & Wooldridge (2008).

3.2.1. Cross-sectional studies. A large number of studies in environmental economics use cross-sectional variation in amenities (e.g., pollution levels or access to parks) to derive the WTP for such amenities. Most of these amenities are not directly traded in a market, so indirect methods are the only possible option to derive preferences. Using a sample of houses in St. Louis, Missouri, Ridker & Henning (1967) were among the first to argue that better air quality capitalizes into housing values. If two houses are identical (same

housing attributes, neighborhood characteristics, distance to urban centers/schools, etc.), except that one is in a polluted area while the other is in a clean area, one would expect that the price of the latter should be higher. Rosen (1974) formalized the concept of hedonic bid functions, whose underlying assumptions are outlined in more detail in Section 2.2.

The big empirical challenge when estimating the marginal WTP is to account for all other factors that might differ between various houses. Smith & Huang (1993, 1995) conducted meta-analyses of hedonic studies that estimate the WTP for air quality. Smith & Huang (1993) found that hedonic studies are successful at establishing that pollution impacts housing values. In their follow-up paper (Smith & Huang 1995), they found that the chosen procedure had an important influence on the estimated marginal WTP for improvements in air quality. When the authors regressed the estimated marginal WTP of each study on characteristics of the study, they found that the number of neighborhood control variables is positive and significant, as are dummies for linear, semilog, and log-linear specifications as well as controls for the time the study was conducted. The mean estimate is five times as large as the median, suggesting that outliers can have a large impact. Although none of these is unique to cross-sectional studies, it shows that great care should be taken in the study design and in determining what it can be used for.

In another meta-study, Boyle & Kiel (2001) found that hedonic pricing studies of air quality more frequently have insignificant coefficients or even coefficients with counterintuitive signs compared with studies that value water quality or the distance to undesirable land use. One possible reason is that water quality measures (e.g., turbidity) and the distance to toxic waste sites are easily observable, unlike common air quality measures, such as ozone, that can be observed only with special instruments.

Given these mixed results and empirical challenges, why are researchers employing them? The main reason may be the lack of alternatives combined with the fact that alternatives often require assumptions of their own. The first advantage of reduced-form cross-sectional studies is that they are more easily understood than some structural models, and it is more evident what source of variation is used to explain the variation in the dependent variable. Second, quasi-experiments are not always available. If they are available, they may measure a different parameter than a cross-sectional study or require assumptions of their own (discussed below). Kuminoff & Pope (2009) highlighted the theoretical reasons why a cross-sectional approach and a quasi-experimental approach may give different estimates. The former measures the marginal WTP at a market equilibrium at a single point in time, whereas the latter usually measures the capitalization of a discrete nonmarginal change where equilibrium prices and quantities adjust. Only under certain conditions, e.g., constant prices and the assumption that the marginal WTP is constant over the change, will the two coincide. Because quasi-experimental approaches usually evaluate larger differences that sometimes span an extended period of time, these assumptions are not always met. Third, a cross section may be the only setting that allows researchers to measure how people adapt to changes in long-term averages (discussed in Section 3.2.2).

3.2.2. Panel studies. A panel setting allows the use of fixed effects to account for all time-invariant factors that impact the decision in an additive way. This method is, hence, less susceptible to omitted variables bias than a cross-sectional approach. The source of identi-

fication is different: One does not rely on the difference in the average of each unit (individuals, households, or spatial units such as zip codes), but instead one relies on deviation from the mean of each unit to identify the parameters of interest. For example, Currie & Neidell (2005) used the variation in pollution levels over time to identify the impact of several pollutants on infant mortality using zip-code fixed effects as well as year and month fixed effects and the interactions of these fixed effects. Such an approach allows average mortality rates to be different between zip codes and relies only on variation within locations over time (net of seasonality effects of overall yearly time trends) to estimate effect of pollution on mortality.

A fixed-effect regression is equivalent to a joint demeaning of the dependent and independent variables. By definition, time-invariant factors do not evolve over time; therefore, subtracting the constant mean subtracts out the additive influence of such variables. The advantage of implicitly accounting for all time-invariant factors, however, comes at a cost: Deviation from means can have distinct economic implications than differences in means. A panel relies on period-to-period variation and is hence inherently a short-term response, whereas the cross-sectional approach uses differences in long-term averages to which units can adapt. The direction of the difference can go either way, depending on how the set of possible responses varies in the short and long term. If the set of possible responses is wider in the long run (e.g., because additional investments are possible that cannot be accomplished in the short term), the long-run impact should be a larger net benefit (or lower cost). This is the Le Chatelier principle—responses in the long term have fewer constraints than in the short term and hence are less harmful or more beneficial. In contrast, within the environmental area, there may also be short-term responses that cannot be sustained indefinitely (as the resources necessary to do so would be depleted) and hence the short-run impact may give higher net benefits. For example, a one-time bacterial infection can easily be fought by using antibiotics, i.e., the response to a short-term infection is to increase the use of antibiotics. However, continued use of antibiotics makes bacteria resistant, so the use of antibiotics on a long-term basis would not be feasible. Similarly, a drought can be mitigated by pumping groundwater, but continued droughts may require more water than the aquifer holds.

Another difference between the short and long term is that individuals are stuck in the former but can re-sort in the latter. Under Tiebout (1956) sorting, people with a high aversion to a nuisance, e.g., proximity to an industrial facility with emissions in the Toxic Release Inventory, sort to neighborhoods that do not have a facility in close proximity. Banzhaf & Walsh (2008) showed that air quality improvements lead to increased population density, as people migrate to the cleaner neighborhood, and increased average income levels. Such sorting implies that, as they sorted, families in clean areas had different marginal WTP for improvements in air quality. Therefore, deviations from mean levels may differ for at least two reasons. First, individuals with high values for clean areas may re-sort into cleaner areas and hence have a different marginal WTP, even if pollution levels were identical. Second, the marginal WTP may be increasing in the pollution level if damages are convex. Under convex damages, the marginal effect depends on the initial pollution level; hence, the marginal WTP is heterogeneous and determined by initial starting values. Although Banzhaf & Walsh (2008) conducted a reduced-form study that estimates the net effect of re-sorting, it highlights why structural approaches (outlined in the first half of this article) that explicitly incorporate such sorting are a useful tool to evaluate nonmarginal changes in amenities.

To avoid possible sorting responses, it is ideal if the variable whose effect the researcher is interested in measuring is exogenously assigned. Quasi-experiments, discussed next, focus on the exogenous assignment of a treatment on individuals. All other initial controls or initial levels should be balanced, i.e., identical between the treatment and control groups.

3.2.3. Quasi-experiments. It is often impossible to conduct a randomized experiment in the economy, but unique events can act like a natural experiment or quasi-experiment. Such events ideally should assign a “treatment” to a subset of the observations, and this assignment should be random. For an example of a quasi-experiment, consider the relationship between wind direction and pollution. Except within dense city centers with skyscrapers, wind direction is exogenously given and varies daily or hourly in most locations. When examining the effects of pollution, one can compare days on which a location is downwind with days on which it is upwind of the pollution source. All characteristics except the pollution level should be balanced between the two samples. Different outcomes between days with various wind directions can hence be attributed to the pollution.

Chay & Greenstone (2003) used changes in pollution that were induced by a recession to estimate the effect of air pollution on infant health. The authors compared the change in infant mortality rates in the third of the counties with the largest reduction in pollution levels to the third of counties with the smallest change in pollution levels. The idea is that the recession induced exogenous pollution reductions; therefore, all other confounding variables should be balanced between the two groups of counties. The authors tested whether various socioeconomic variables are balanced between these subsamples and found only a small difference in income—so small that it cannot explain the observed differences in the measured response. The estimated relationship between pollution and infant health is much larger and more robust than estimates obtained in a cross section.

Davis (2004) examined the effect a sudden spike in pediatric cancer cases in a town in Nevada had on housing values. The event is the sudden occurrence of an unusually larger number of such cases within a limited time window. He used two towns in Nevada that have very similar pre-event trends in housing prices. The town with the sudden number of cancer cases is the treatment town, whereas the other with no occurrences acts like a control. Because the occurrence of the cancer cases started and ceased unexpectedly and suddenly, this method was similar to assigning a treatment to one town but not to the other. Davis used, among other things, a difference-in-difference estimator by first constructing the difference in housing prices before and after the event in both the treatment and control village. The second difference, or difference-in-difference estimator, takes the difference between the change in the housing price in the treatment village where the cancer cases occurred and compares it to the difference in the control village. The advantage of a difference-in-difference approach is that, by taking the second difference, the analysis subtracts the change in housing prices observed in the control village, thus removing overall price movements that are common throughout Nevada. Davis found a strong and statistically significant reduction in housing prices of approximately 15% in the treatment town compared with the control town following the discovery of the cancer cases. Similar results are obtained in a cross-sectional analysis of housing prices.

Fowlie & Perloff (2008) used the random assignment of firms to one of two compliance cycles of the RECLAIM market for NO_x emissions in Southern California to test whether pollution outcomes are independent of initial permit allocation, as predicted under the Coase theorem. Firms receive facility-specific permit allocation schemes that

decreased in a stepwise function over time. Permits are valid for 12-month periods and may be traded between firms. However, regulations prohibit banking, i.e., the trading of permits across time periods. The quasi-experimental design is possible because of the way the permits were issued. Firms were randomly assigned to one of two 12-month cycles, one ending in January and the other ending in July. Since firms are not allowed to shift permits between time periods, the declining permit structure was exogenously given by a random draw. The authors found that the null hypothesis that emissions are independent of initial permit allocation cannot be rejected. The analysis establishes this result convincingly as it relies on exogenous variation in the timing of the permit profile, which is independent of firm characteristics that influence overall permit allocation. Such randomization ensures that the variation is orthogonal to other possible confounding factors.

Another strand of quasi-experimental research designs is the regression discontinuity framework. The idea is to look for a treatment that switches on when a critical threshold value is reached. For this approach to work, there must not be endogenous “bunching” at the threshold level, where individuals can choose whether they will be above or below it. A continuous exogenous distribution around the threshold ensures that an observation just above and just below the threshold should be comparable, except that one receives the treatment while the other does not. For example, Urquiola & Verhoogen (2009) showed that there is bunching around the class size cutoff in schools. Thus, a regression discontinuity design in class size is problematic.

Auffhammer & Kellogg (2009) applied the regression discontinuity method to examine whether gasoline content regulation has an effect on air quality. The authors used the fact that various regulations in the United States are binding for only parts of the year, mostly the summer months as higher temperatures are conducive to ozone formation. The authors then estimated whether days just before and after the phase-in have different pollution levels. These levels should be comparable, especially when days before and after the phase-in each is averaged over several years, as idiosyncratic shocks are averaged out. The authors found that most gasoline content regulations that regulate volatile organic compounds (VOC) do not reduce ozone formation, as these regulations specify only a limit on the total quantity of VOCs emitted, but not which type. Not surprisingly, firms meet the regulation by eliminating the cheapest source of VOCs, which has little or no impact on ozone formation. In contrast, a California law that postulates which VOCs have to be removed does lead to significant improvements in air quality.

A regression discontinuity approach is similar in spirit to an event study approach in finance, where abnormal stock market returns are linked to the event in question. In the event study, the discontinuity does not happen along a state variable (e.g., pollution levels), but rather in time. New information is suddenly made available and financial markets react to the release of this new information. Arbitrage requires that such new information be incorporated immediately. Hamilton (1995) found that firms with the largest emissions, as reported in the Toxic Release Inventory, had negative stock market returns after the information was made initially public. The point estimate for the effect of Toxic Release Inventory emissions is larger for the event date than for the week following the event, suggesting that new information is immediately incorporated in efficient markets. Such methodology is suited for cases in which it is clear when new information becomes available, e.g., sudden catastrophes or unexpected changes. It is important to be able to identify the temporal window when the stock market recognizes the information as “new.”

The effect of sudden changes on firms' values has been examined in a wide variety of settings. Fisman (2001) showed that firms connected to Indonesia's former dictator Suharto had abnormal returns when concerns surfaced about his health. Similarly, Jayachandran (2006) examined the event when Senator Jeffords unexpectedly switched party affiliations and tipped the U.S. Senate majority from Republican to Democratic. Firms that predominantly donated to Republicans had negative returns, while firms with larger donations to Democrats had positive returns. Lange & Linn (2008) used the uncertainty surrounding the recount of the 2000 U.S. Presidential election in Florida to estimate the cost of the New Source Review standard under the Clean Air Act. On the one hand, it was believed that Gore would interpret the rule more stringently, which would lead to higher costs for industries with regulated boilers. On the other hand, Bush was expected to interpret the rule more loosely. The sudden back-and-forth between the Florida and U.S. Supreme Court rulings changed the probability that each candidate would capture Florida and win the overall election. As a result, the profit of companies would shift with the predicted winner as the regulation would be more or less severe.

3.2.4. Randomization: field experiments. Whereas quasi-experiments examine events that split the sample into treatment and control groups that are arguably random, field experiments go one step further and deliberately randomize an experiment. By design, a certain fraction of the overall population actually receives one out of a set of treatment options. The most famous examples are medical trials, where a fraction of the patients receive a new drug whose effect is tested and the control sample receives a placebo (usually a sugar pill). The outcome of the sample that received the treatment is then compared with the sample that did not.

In the economic literature, field experiments are prevalent in various other disciplines, notably, development economics and the economics of education. The major advantage of randomized experiments is that, by design, the treatment is uncorrelated with any other controls. As a result, the treatment effect is clearly identified. The statistical tools required in such an analysis are simple and often contain only t-tests to examine whether the means of the treatment and control group are different. Heterogeneous treatment effects can be examined by interacting the treatment variable with other controls. For example, Landry et al. (2006) examined the effects of various fund-raising mechanisms. They first argued that lotteries should raise more money than voluntary donations. Under the former, the donor not only gets the "joy of giving," but also has the possibility of a private payoff. They tested this proposition in a field experiment in North Carolina in which 5000 potential donors were approached with a randomly chosen donation mechanism. As hypothesized, the lottery resulted in higher average donations than voluntary giving. But the authors did not stop at the difference between various donation mechanisms. They also interacted the donation amount with various measures of the beauty of the solicitors to test whether physical attractiveness has an impact on giving. A one standard deviation increase in the beauty measure increased average giving by about the same as the lottery incentive, compared with voluntary donations.

Field experiments allow a researcher to observe human behaviors and attitudes that are difficult to establish in a cross section. For example, there is an ongoing debate in the environmental justice literature as to whether plants locate disproportionately in minority neighborhoods, or whether minorities, which on average are poorer, tend to move to areas with more plants because rents are lower (Wolverton 2008). Establishing a causal link by

comparing neighborhoods is very difficult. List (2004) used interactions between various ethnic groups in a field experiment where individuals trade baseball cards to test discrimination. The advantage is that the moderator can influence the racial composition of the buyer and seller. List (2004) found that trading partners from minority groups tend to get lower offers, but a subsequent analysis revealed that these lower offers are not due to animus or taste-based preferences toward the group, but rather statistical discrimination is the cause. In other words, market participants aim at maximizing profit (irrespective of their counterpart's ethnic group) and use race as an indicator of private reservation values.

Although randomized experiments offer a clear form of identification, the researcher also has to ensure that the assigned treatment was eventually implemented. Sometimes, subjects cannot be forced to undergo treatment but can only be offered the option to be treated. A subject's decision to be treated is endogenous to the subject and is based on unobservable characteristics. The researcher has to account for such selection. The intend-to-treat literature addresses such issues.

Moreover, it is important to ensure that there are no spillover effects from the treatment to the control group. This is particularly true if subjects are randomly given additional information or monetary incentives that they could share with their neighbors or relatives in the control group. Such behavior will dilute the measured effect of the treatment, as part of the control group indirectly receives the treatment as well. For example, Ferraro (2009) convinced a water utility in Georgia to add information randomly about a household's water use (compared with the average user) to the water bill. He found that simply providing such information changes the water use of the household, but he did not detect a change in neighbors' consumption, so spillover effects seem not to be a major issue.

Finally, Deaton (2009) recently emphasized that experimental (or quasi-experimental) approaches have sometimes given the false illusion of exogeneity and rest on other assumptions. Moreover, he emphasized that the randomized field experiments have focused too much on whether a particular policy works, and not why it works.

3.2.5. Randomization: lab experiments. Lab experiments offer a cheaper alternative to field experiments, which are often expensive and administratively cumbersome to implement. Moreover, a lab experiment can create a payoff structure that may not be implemented in the field owing to ethical concerns. For example, Shogren et al. (1994) examined the differences between WTP and willingness to accept (WTA) for market goods as well as nonmarket goods with few substitutes, e.g., health risks. In a laboratory experiment, people were first given a market good (candy) and some income. In an auction, WTP and WTA converged when the same experiment was repeated. Subjects were then given \$15 and a free lunch from a local supplier that had a certain percentage of being contaminated with food pathogens. Participants WTP and WTA were subsequently tested for a scanned "safe" lunch with lower probability of food poisoning, and the two measures did not converge over time.

Lab experiments can also be used to test specific implementation issues of various policy settings and whether individuals react as predicted by economic theory. For example, the CAAA of 1990 require the Environmental Protection Agency to conduct annual auctions of emission allowances. The unique feature of this auction rule is that sellers are ranked according to increasing asking prices while buyers are ranked according to decreasing bid prices. The seller with the lowest asking price then receives the bid of the

highest bid. There is an incentive to underbid in order to be paired with the highest bidder and thereby receive a higher price. Cason (1995) examined the opposite situation in a laboratory experiment in which buyers with the highest bid pay the lowest asking price. As predicted by theory, buyers bid above their true value, driving up the equilibrium price to inefficient levels.

Although experiments can be tailored to various setups, potential downsides also exist. First, experiments are usually conducted in an artificial setting, often administered via computers, and we learn about how humans cooperate and behave given these recreated incentive structures. Second, subjects are predominantly students who participate in an experiment and hence may not be representative of the overall population. Third, Levitt & List (2007) argued that responses in lab experiments are governed by factors other than monetary calculations, e.g., the specific context, social stigma, moral and ethical considerations, as well as the stakes of the game.

3.2.6. Reduced-form studies: possible challenges. The key strength of reduced-form studies is a clear and transparent identification of a parameter of interest without structural assumptions about feedback loops. Possible challenges are that both long-term feedback effects and spillovers can be important for welfare analyses.

Cohort studies can sometimes address long-term effects after people are able to reoptimize. Such studies rely on an event where an entire cohort of people is subjected to a shock. The cohort's eventual outcome revealed several years later is contrasted to the outcomes of previous and later cohorts. For example, Almond (2006) estimated the long-term effects of the big influenza epidemic of 1918 by contrasting cohorts that were in utero in 1918 to cohorts right before and after the epidemic. He used the 1960–1980 Censuses to examine long-term effects and found that the exposed cohort had higher rates of disability, unemployment, and various other factors even at a much later stage in life.

The second problem of spillover effects is more severe. Treatment-control experiments usually assume that the treatment has no effect on the control group. Imbens & Wooldridge (2008, p. 9) argued that, although this seems justifiable for medical research, interactions between the treatment and control group can be a serious concern in economic studies. For example, a labor market treatment may have spillover effects for the control group by impacting the market-clearing wage rate, even if a person is in the control group. The authors cited some recent advances that see these interactions not as a problem but as an interesting objective of the study. For example, Kling et al. (2007) examined the outcomes of individuals five years after they randomly received housing vouchers, giving them time to re-sort. Imbens & Wooldridge (2008) conceded that “[m]any identification and inferential questions remain unanswered in this literature” (p. 10). This line of research should be expanded in the future.

3.3. Summary

Reduced-form studies are best equipped to identify key high-level parameters of interest. The major advantage is that the identification strategy often requires fewer assumptions and is easier to grasp than structural models with all their embodied feedback loops while trying to estimate underlying primitives. We have discussed how randomization simplifies the task at hand for an econometrician, as all covariates are by design balanced between the treatment and control group. Such randomized experiments offer the most compelling

evidence to establish a causal argument between a treatment and an outcome, yet these studies require assumptions of their own. Field experiments offer the most realistic setting, yet some problems involving pollution and health can be simulated only in a hypothetical lab setting for ethical reasons. Cross-sectional studies sometimes offer the best possible setting to test how individuals adapt to long-term differences. For these reasons, both quasi-experiments and cross-sectional studies can give viable insights. As described above, there are theoretical reasons why quasi-experiments that rely on significant nonmarginal changes can give estimates of the marginal WTP that differ from those obtained by cross-sectional studies.

Returning to the article by Freedman (1991) and his discussion of the statistical evidence surrounding the hypothesis that cholera is a waterborne disease, the most convincing fact was that the researcher employed “shoe leather,” i.e., he tested various sources of variation and models and found them to agree.

4. CONCLUSIONS

In this article, we have discussed both structural and reduced-form models as well as their strengths and weaknesses. These strengths and weaknesses are not unique to environmental and resource economics; instead, they are common to other subfields in economics. Structural models can allow for feedback loops that are difficult, if not impossible, to implement in a reduced-form setting. A complete welfare analyses may require these structural feedback loops. However, reduced-form studies can offer a clear identification of key high-level response parameters of interest and are better suited at establishing causality. They can also be used to test predictions from economic theory using data-driven approaches that are less dependent on modeling assumptions. We believe both approaches have their own unique advantages. For example, in the case of air quality, reduced-form studies have been used to establish the link between air quality and infant health using variation induced by a recession (Chay & Greenstone 2003) or using panel data that link health outcomes to variation in pollution taking into account location fixed effects as well as temporal fixed effects (Currie & Neidell 2005). Both studies find a robust and consistent relationship between reduction in pollution and infant mortality. The latter study uses more recent data when air quality levels had improved further compared with the first study that uses data in the early 1980s. However, to value the WTP for large reduction in air quality, re-sorting of constituents into cleaner or dirtier neighborhoods can become important. The structural paper by Sieg et al. (2004) models the ability of people in the Los Angeles area to relocate in response to a large reduction in pollution levels, and it shows how relocation would impact estimates of their marginal WTP for that improvement. The horizontal model used by Bayer et al. (2009) alternatively shows how stickiness in the sorting process can lead to a downward bias in WTP as measured by a simple hedonic model.

Generally, the most convincing evidence stems from contrasting various approaches instead of limiting ourselves to only one approach and, in the process, enabling us to discuss why these approaches agree or disagree.

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